LECTURE 1

Our lecture will be online

But exams will be face to face,

Exam weeks 7th November-14th November

Exam will be multiplice choice:

0.1 EVALUATING A POLYNOMIAL

What is the best way to evaluate

$$P(x) = 2x^4 + 3x^3 - 3x^2 + 5x - 1,$$

say, at $x = 1/2$? Assume that the coefficients of the polynomial and the number 1/2 are stored in memory, and try to minimize the number of additions and multiplications required

$P(\frac{1}{2}) = 2\cdot(\frac{1}{2})^4 + 3\cdot(\frac{1}{2})^3 - 3\cdot(\frac{1}{2})^2 + 5(\frac{1}{2}) - 1$

In order evaluate $P(\frac{1}{2})$, how many multiplication and addition, should computer do?

$P(\frac{1}{2}) = 2*(\frac{1}{2})*(\frac{1}{2})*(\frac{1}{2})*(\frac{1}{2}) + 3*(\frac{1}{2})*(\frac{1}{2})*(\frac{1}{2}) - 3*(\frac{1}{2})*(\frac{1}{2}) + 5*(\frac{1}{2}) - 1$

multiplication number=10, addition. number=4

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METHOD 1 The first and most straightforward approach is

$$P\left(\frac{1}{2}\right) = 2 * \frac{1}{2} * \frac{1}{2} * \frac{1}{2} * \frac{1}{2} + 3 * \frac{1}{2} * \frac{1}{2} * \frac{1}{2} - 3 * \frac{1}{2} * \frac{1}{2} + 5 * \frac{1}{2} - 1 = \frac{5}{4}. \quad (0.1)$$

The number of multiplications required is 10, together with 4 additions. Two of the additions are actually subtractions, but because subtraction can be viewed as adding a negative stored number, we will not worry about the difference.

There surely is a better way than (0.1). Effort is being duplicated—operations can be saved by eliminating the repeated multiplication by the input 1/2. A better strategy is to first compute $(1/2)^4$, storing partial products as we go. That leads to the following method:

Number of total operations in the first method is 10+4=14

METHOD 2 Find the powers of the input number $x = 1/2$ first, and store them for future use:

$$\frac{1}{2} * \frac{1}{2} = \left(\frac{1}{2}\right)^2$$
$$\left(\frac{1}{2}\right)^2 * \frac{1}{2} = \left(\frac{1}{2}\right)^3$$
$$\left(\frac{1}{2}\right)^3 * \frac{1}{2} = \left(\frac{1}{2}\right)^4.$$

Now we can add up the terms:

$$P\left(\frac{1}{2}\right) = 2 * \left(\frac{1}{2}\right)^4 + 3 * \left(\frac{1}{2}\right)^3 - 3 * \left(\frac{1}{2}\right)^2 + 5 * \frac{1}{2} - 1 = \frac{5}{4}.$$

There are now 3 multiplications of 1/2, along with 4 other multiplications. Counting up,

There are now 3 multiplications of 1/2, along with 4 other multiplications. Counting up,
we have reduced to 7 multiplications, with the same 4 additions. Is the reduction from 14 to 11 operations a significant improvement? If there is only one evaluation to be done, then probably not. Whether Method 1 or Method 2 is used, the answer will be available before you can lift your fingers from the computer keyboard. However, suppose the polynomial needs to be evaluated at different inputs x several times per second. Then the difference may be crucial to getting the information when it is needed.

Is this the best we can do for a degree 4 polynomial? It may be hard to imagine that we can eliminate three more operations, but we can. The best elementary method is the following one:

$$P(x) = 2x^4 + 3x^3 - 3x^2 + 5x - 1,$$

$$= x(2x^3 + 3x^2 - 3x + 5) - 1$$
$$= x(x(2x^2 + 3x - 3) + 5) - 1$$
$$= x(x(x(2x + 3) - 3) + 5) - 1$$

How many multiplications and additions you need to evaluate $P(\frac{1}{2})$?

$= x\{x(x(2x+3)-3)+5\}-1$

How many multiplications and additions you need to evaluate $P(\frac{1}{2})$?

$P(\frac{1}{2}) = \frac{1}{2}*\left(\frac{1}{2}*\left(\frac{1}{2}*\left(2*\frac{1}{2}+3\right)-3\right)+5\right)-1$

of multiplication: 4

of addition: 4

Total number: 8

METHOD 3 (Nested Multiplication) Rewrite the polynomial so that it can be evaluated from the inside out:

$$\begin{aligned} P(x) &= -1 + x(5 - 3x + 3x^2 + 2x^3) \\ &= -1 + x(5 + x(-3 + 3x + 2x^2)) \\ &= -1 + x(5 + x(-3 + x(3 + 2x))) \\ &= -1 + x * (5 + x * (-3 + x * (3 + x * 2))). \end{aligned} \tag{0.2}$$

Here the polynomial is written backwards, and powers of x are factored out of the rest of the polynomial. Once you can see to write it this way—no computation is required to do the rewriting—the coefficients are unchanged. Now evaluate from the inside out:

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This method, called **nested multiplication** or **Horner's method**, evaluates the polynomial in 4 multiplications and 4 additions. A general degree d polynomial can be evaluated in d multiplications and d additions. Nested multiplication is closely related to synthetic division of polynomial arithmetic.

▶ **EXAMPLE 0.1** Find an efficient method for evaluating the polynomial $P(x) = 4x^5 + 7x^8 - 3x^{11} + 2x^{14}$.

use nested multiplication method.

$P(x) = x^5(4 + 7x^3 - 3x^6 + 2x^9)$

$= x^5(4 + x^3(7 - 3x^3 + 2x^6))$

$= x^5 * (4 + x^3 * (7 + x^3 * (-3 + 2x^3)))$

How many multiplications: 7

How many additions: 3

We need to calculate

$x^5 = x*x*x*x*x$

4 operations

$x^2 = x*x$ 1 operation save

$x^3 = x*x^2$ 2 operation save

in 2 operation we calculate x^2 and x^3

$x^5 = x^2 * x^3$, 3 multiplication

▸ **EXAMPLE 0.1** Find an efficient method for evaluating the polynomial $P(x) = 4x^5 + 7x^8 - 3x^{11} + 2x^{14}$.

Some rewriting of the polynomial may help reduce the computational effort required for evaluation. The idea is to factor x^5 from each term and write as a polynomial in the quantity x^3:

$$\begin{aligned} P(x) &= x^5(4 + 7x^3 - 3x^6 + 2x^9) \\ &= x^5 * (4 + x^3 * (7 + x^3 * (-3 + x^3 * (2)))). \end{aligned}$$

For each input x, we need to calculate $x * x = x^2$, $x * x^2 = x^3$, and $x^2 * x^3 = x^5$ first. These three multiplications, combined with the multiplication of x^5, and the three multiplications and three additions from the degree 3 polynomial in the quantity x^3 give the total operation count of 7 multiplies and 3 adds per evaluation. ◂

Exercises

1. Rewrite the following polynomials in nested form. Evaluate with and without nested form at $x = 1/3$.
   (a) $P(x) = 6x^4 + x^3 + 5x^2 + x + 1$
   (b) $P(x) = -3x^4 + 4x^3 + 5x^2 - 5x + 1$
   (c) $P(x) = 2x^4 + x^3 - x^2 + 1$
2. Rewrite the following polynomials in nested form and evaluate at $x = -1/2$:
   (a) $P(x) = 6x^3 - 2x^2 - 3x + 7$
   (b) $P(x) = 8x^5 - x^4 - 3x^3 + x^2 - 3x + 1$
   (c) $P(x) = 4x^6 - 2x^4 - 2x + 4$
3. Evaluate $P(x) = x^6 - 4x^4 + 2x^2 + 1$ at $x = 1/2$ by considering $P(x)$ as a polynomial in x^2 and using nested multiplication.

BINARY NUMBERS

In preparation for the detailed study of computer arithmetic in the next section, we need to understand the binary number system. Decimal numbers are converted from base 10 to base 2 in order to store numbers on a computer and to simplify computer operations like addition and multiplication. To give output in decimal notation, the process is reversed. In this section, we discuss ways to convert between decimal and binary numbers.

Binary numbers are expressed as

$$\ldots b_2 b_1 b_0 . b_{-1} b_{-2} \ldots ,$$

where each binary digit, or **bit**, is 0 or 1. The base 10 equivalent to the number is

$$\ldots b_2 2^2 + b_1 2^1 + b_0 2^0 + b_{-1} 2^{-1} + b_{-2} 2^{-2} \ldots .$$

where each binary digit, or **bit**, is 0 or 1. The base 10 equivalent to the number is

$$\ldots b_2 2^2 + b_1 2^1 + b_0 2^0 + b_{-1} 2^{-1} + b_{-2} 2^{-2} \ldots .$$

like in base 10:

$$2534.23 = 2\times10^3 + 5\times10^2 + 3\times10^1 + 4\times10^0 + 2\times10^{-1} + 3\times10^{-2}$$

in base 2:

$$(110101.101)_2 = 1\times2^5 + 1\times2^4 + 0\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 + 1\times2^{-1} + 0\times2^{-2} + 1\times2^{-3}$$
$$= 32 + 16 + 0 + 4 + 0 + 1 + \frac{1}{2} + 0 + \frac{1}{8}$$
$$= 53 + 0.5 + 0.125 = 53.625$$

To convert the number, for example 53.7 to base 2.
First separate integer and fractional parts of 53.7

$$53.7 = 53 + 0.7$$

What is 53 in base 2?

$53 = (110101)_2$

53 | 2	
52 | 26 | 2	
1 26 | 13 | 2	
0 12 | 6 | 2	
1 6 | 3 | 2	
0 2 | 1	
1	

0.2.1 Decimal to binary

The decimal number 53 will be represented as $(53)_{10}$ to emphasize that it is to be interpreted as base 10. To convert to binary, it is simplest to break the number into integer and fractional parts and convert each part separately. For the number $(53.7)_{10} = (53)_{10} + (0.7)_{10}$, we will convert each part to binary and combine the results.

What is 0.7 in base 2?

$0.7 \times 2 = 1.4 \rightarrow$ take 1
$0.4 \times 2 = 0.8 \rightarrow$ take 0
$0.8 \times 2 = 1.6 \rightarrow$ take 1
$0.6 \times 2 = 1.2 \rightarrow$ take 1
$0.2 \times 2 = 0.4 \rightarrow$ take 0
$0.4 \times 2 = 0.8 \rightarrow 0$

It is repeating again

So $0.7 = (0.\overline{10110})_2$

Then $53.7 = (110101.\overline{10110})_2$

0.2.1 Decimal to binary

The decimal number 53 will be represented as $(53)_{10}$ to emphasize that it is to be interpreted as base 10. To convert to binary, it is simplest to break the number into integer and fractional parts and convert each part separately. For the number $(53.7)_{10} = (53)_{10} + (0.7)_{10}$, we will convert each part to binary and combine the results.

Integer part. Convert decimal integers to binary by dividing by 2 successively and recording the remainders. The remainders, 0 or 1, are recorded by starting at the decimal point (or more accurately, **radix**) and moving away (to the left). For $(53)_{10}$, we would have

$$\begin{aligned} 53 \div 2 &= 26 \text{ R } 1 \\ 26 \div 2 &= 13 \text{ R } 0 \\ 13 \div 2 &= \ 6 \text{ R } 1 \\ 6 \div 2 &= \ 3 \text{ R } 0 \\ 3 \div 2 &= \ 1 \text{ R } 1 \end{aligned}$$

Therefore, the base 10 number 53 can be written in bits as 110101, denoted as $(53)_{10} = (110101.)_2$. Checking the result, we have $110101 = 2^5 + 2^4 + 2^2 + 2^0 = 32 + 16 + 4 + 1 = 53$.

Fractional part. Convert $(0.7)_{10}$ to binary by reversing the preceding steps. Multiply by 2 successively and record the integer parts, moving away from the decimal point to the right.

$$\begin{aligned} .7 \times 2 &= .4 + 1 \\ .4 \times 2 &= .8 + 0 \\ .8 \times 2 &= .6 + 1 \\ .6 \times 2 &= .2 + 1 \\ .2 \times 2 &= .4 + 0 \\ .4 \times 2 &= .8 + 0 \\ &\vdots \end{aligned}$$

Notice that the process repeats after four steps and will repeat indefinitely exactly the same way. Therefore,

$$(0.7)_{10} = (.1011001100110\ldots)_2 = (.1\overline{0110})_2,$$

where overbar notation is used to denote infinitely repeated bits. Putting the two parts together, we conclude that

$$(53.7)_{10} = (110101.1\overline{0110})_2.$$

0.2.2 Binary to decimal

To convert a binary number to decimal, it is again best to separate into integer and fractional parts.

Integer part. Simply add up powers of 2 as we did before. The binary number $(10101)_2$ is simply $1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 = (21)_{10}$.

Fractional part. If the fractional part is finite (a terminating base 2 expansion), proceed the same way. For example,

$$(.1011)_2 = \frac{1}{2} + \frac{1}{8} + \frac{1}{16} = \left(\frac{11}{16}\right)_{10}.$$

The only complication arises when the fractional part is not a finite base 2 expansion. Converting an infinitely repeating binary expansion to a decimal fraction can be done in several ways. Perhaps the simplest way is to use the shift property of multiplication by 2.

For example, suppose $x = (0.\overline{1011})_2$ is to be converted to decimal. Multiply x by 2^4, which shifts 4 places to the left in binary. Then subtract the original x:

$$\begin{aligned} 2^4 x &= 1011.\overline{1011} \\ x &= 0000.\overline{1011}. \end{aligned}$$

Subtracting yields

$$(2^4 - 1)x = (1011)_2 = (11)_{10}.$$

Then solve for x to find $x = (.\overline{1011})_2 = 11/15$ in base 10.

As another example, assume that the fractional part does not immediately repeat, as in $x = .10\overline{101}$. Multiplying by 2^2 shifts to $y = 2^2 x = 10.\overline{101}$. The fractional part of y, call it $z = .\overline{101}$, is calculated as before:

$$\begin{aligned} 2^3 z &= 101.\overline{101} \\ z &= 000.\overline{101}. \end{aligned}$$

Therefore, $7z = 5$, and $y = 2 + 5/7$, $x = 2^{-2} y = 19/28$ in base 10. It is a good exercise to check this result by converting $19/28$ to binary and comparing to the original x.

Binary numbers are the building blocks of machine computations, but they turn out to be long and unwieldy for humans to interpret. It is useful to use base 16 at times just to present numbers more easily. **Hexadecimal numbers** are represented by the 16 numerals $0, 1, 2, \ldots, 9, A, B, C, D, E, F$. Each hex number can be represented by 4 bits. Thus $(1)_{16} = (0001)_2$, $(8)_{16} = (1000)_2$, and $(F)_{16} = (1111)_2 = (15)_{10}$. In the next section, MATLAB's `format hex` for representing machine numbers will be described.

Integ

$(10101)_2 = 1\times2^4 + 0\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0$

$= 16+0+4+0+1 = 21$

$(0.1011)_2 = 1\times2^{-1} + 0\times2^{-2} + 1\times2^{-3} + 1\times2^{-4}$

$= \frac{1}{2} + 0 + \frac{1}{8} + \frac{1}{16} = \frac{11}{16}$

(8) (2)

$(0.\overline{1011})_2 = (0.1011011011011\ldots)$

$x = 57.281365$

$10^4 x = 572813.65$

or $x = 57.\overline{2813}$

$10^4 x = 572813.\overline{2813}$

For example, suppose $x = (0.\overline{1011})_2$ is to be converted to decimal. l

$(0.\overline{1011})_2 = (\ ?\ )_{10}$

Soln: let $x = (0.\overline{1011})_2$ (I)

$2^4 x = (1011.\overline{1011})_2$ (II)

(II) − (I) $= 16x - x = 15x = (1011)_2$

$(1011)_2 = 8+0+2+1 = 11$

$15x = 11$ $x = \frac{11}{15}$, that is: $(0.\overline{1011})_2 = \frac{11}{15}$

Another example, convert the number $x = (0.10\overline{101})_2$ in decimal number.

Soln: So $2^2 x = (10.\overline{101})_2 = (10)_2 + (0.\overline{101})_2$

$4x = 2 + 0 + (0.\overline{101})_2$

we need to convert $y = (0.\overline{101})_2$ as before:

$\Rightarrow 2^3 y = (101.\overline{101})_2$

$y = (0.\overline{101})_2$

$7y = (101)_2 = 4 + 0 + 1 = 5$ so

$y = (0.\overline{101})_2 = 5/7$

Hence $4x = 2 + \frac{5}{7} = \frac{19}{7}$ thus $x = \frac{19}{28} = (0.10\overline{101})_2$

Exercises

1. Find the binary representation of the base 10 integers. (a) 64 (b) 17 (c) 79 (d) 227
2. Find the binary representation of the base 10 numbers. (a) 1/8 (b) 7/8 (c) 35/16 (d) 31/64
3. Convert the following base 10 numbers to binary. Use overbar notation for nonterminating binary numbers. (a) 10.5 (b) 1/3 (c) 5/7 (d) 12.8 (e) 55.4 (f) 0.1
4. Convert the following base 10 numbers to binary. (a) 11.25 (b) 2/3 (c) 3/5 (d) 3.2 (e) 30.6 (f) 99.9
5. Find the first 15 bits in the binary representation of π.
6. Find the first 15 bits in the binary representation of e.
7. Convert the following binary numbers to base 10: (a) 1010101 (b) 1011.101 (c) $10111.\overline{01}$ (d) $110.\overline{10}$ (e) $10.\overline{110}$ (f) $110.1\overline{101}$ (g) $10.010\overline{1101}$ (h) $111.\overline{1}$
8. Convert the following binary numbers to base 10: (a) 11011 (b) 110111.001 (c) $111.\overline{001}$ (d) $1010.\overline{01}$ (e) $10111.\overline{10101}$ (f) $1111.010\overline{0001}$